Stage 3 — Check Your Understanding

Q1. The effective surface (Proposition 6)

Gaussian profile: psi(s) = exp(-s²/epsilon²)/(sqrt(pi) epsilon).

I = integral[0,infinity] s psi(s) ds = (1/(sqrt(pi) epsilon)) integral[0,infinity] s exp(-s²/epsilon²) ds.

Let t = s²/epsilon², so dt = 2s ds/epsilon² and s ds = (epsilon²/2)dt.

I = epsilon/(2 sqrt(pi)) integral[0,infinity] exp(-t)dt = epsilon/(2 sqrt(pi)).

Boundary-layer solution: u(z) = A + Bz - (tau/mu) Psi(z). The no-slip condition at blob centres u(0)=0 and Psi(0)=0 set A=0.

For |z| >> epsilon: Psi(z) ~ |z|/2 - epsilon/(2 sqrt(pi)).

The closed interior circulation has slip velocity O(tau epsilon/mu) over length R, giving interior slope B + tau/(2mu) = O(tau epsilon/(mu R)), or O(epsilon/R) relative to outer shear.

Hence u_out(z) ~ -(tau/mu)[z - epsilon/(2 sqrt(pi))], which vanishes at z = epsilon/(2 sqrt(pi)), not z=0.

Effective radius: a_eff = R + epsilon/(2 sqrt(pi)) + O(epsilon²/R).

Q2. Why h = epsilon

Ripple from Eq. (5.5): 2 exp[-pi²(epsilon/h)²].

epsilon/h	Substitution	Ripple
0.5	2 exp(-pi²/4)	0.1696
1.0	2 exp(-pi²)	1.0345 x 10^-4
1.5	2 exp(-2.25 pi²)	4.54 x 10^-10

Increasing epsilon/h suppresses leakage by making blobs overlap, but excessive overlap makes the matrix ill-conditioned: its condition number approaches 10^16, near double-precision limits. We stop at h=epsilon because leakage is already 3.3 x 10^-7, the condition number is 5.4 x 10^6, and k varies by less than 2 x 10^-5 relatively once epsilon/h >= 0.76.

Q3. Richardson extrapolation

Assume k(epsilon) = k + c epsilon² and k(2epsilon) = k + 4c epsilon².

Multiply the first by four: 4k(epsilon) = 4k + 4c epsilon².

Subtract the second: 4k(epsilon) - k(2epsilon) = 3k.

Therefore k_R = [4k(epsilon) - k(2epsilon)]/3.

For surface blobs, the error is first order: k(epsilon) = k + c epsilon + O(epsilon²). The second-order weights leave [4k(epsilon)-k(2epsilon)]/3 = k + (2/3)c epsilon + O(epsilon²).

First-order cancellation instead requires k_R^(1) = 2k(epsilon) - k(2epsilon), leaving O(epsilon²) error.

Q4. Reading the random-array results

(a) Supported. At alpha=0.05, random/square = 1.393 with 95% interval [1.322, 1.469], entirely above 1.

(b) Not statistically established. At alpha=0.3, the ratio is 0.944 with 95% interval [0.883, 1.007], which includes 1.

(c) Not supported. At alpha=0.3 and CV=0.5, the ratio after r.m.s.-radius scaling is 1.170 with 95% interval [1.030, 1.317], excluding 1 and showing residual polydispersity.

An interval containing 1 means the data cannot distinguish the ratio from 1 at the stated confidence level; it does not establish equality.